DATA SCIENCE INTERNSHIP

TASK 1

COVID-19 Data Analysis Using Python

Project Type: Data Analysis and Visualization
Language: Python
Libraries: pandas, Matplotlib

Beginner-friendly project guide with objective, complete code, explanation, sample output, instructions, and report.

1. Project Objective

The aim of this project is to explore COVID-19 case, death, and recovery data, summarize worldwide trends, calculate daily changes, and create charts that help communicate the data.

2. Tools Required

• Python 3

• pandas for loading and processing CSV files

• Matplotlib for charts

• Jupyter Notebook, Google Colab, or a Python IDE with internet access

3. Step-by-Step Instructions

1. Open Google Colab or Jupyter Notebook.

2. Install libraries if needed: `pip install pandas matplotlib`.

3. Copy the complete code below into a notebook cell or save it as `covid19_analysis.py`.

4. Run the code with an internet connection. It downloads public Johns Hopkins CSSE historical time-series CSV files.

5. The script creates summary CSV files and PNG charts in the current working folder.

4. Complete Python Code

```
# Task 1: COVID-19 Data Analysis Using Python
# Public historical data from Johns Hopkins CSSE

import pandas as pd
import matplotlib.pyplot as plt

BASE_URL = (
    "https://raw.githubusercontent.com/CSSEGISandData/"
    "COVID-19/master/csse_covid_19_data/"
    "csse_covid_19_time_series/"
)

files = {
    "confirmed": BASE_URL + "time_series_covid19_confirmed_global.csv",
    "deaths": BASE_URL + "time_series_covid19_deaths_global.csv",
    "recovered": BASE_URL + "time_series_covid19_recovered_global.csv",
}

def load_and_aggregate(url, name):
    """Load a time-series CSV and sum daily values across all locations."""
    df = pd.read_csv(url)
    # The first four columns describe location; remaining columns are dates.
    date_columns = df.columns[4:]
    global_daily = df[date_columns].sum(axis=0)
    global_daily.index = pd.to_datetime(global_daily.index, errors="coerce")
    global_daily = global_daily[~global_daily.index.isna()]
    global_daily = global_daily.sort_index()
    global_daily.name = name
    return global_daily

confirmed = load_and_aggregate(files["confirmed"], "Confirmed")
deaths = load_and_aggregate(files["deaths"], "Deaths")
recovered = load_and_aggregate(files["recovered"], "Recovered")

# Combine the time series into one table
data = pd.concat([confirmed, deaths, recovered], axis=1).fillna(0)
data = data.sort_index()

# Calculate daily increases from cumulative totals.
# Some source revisions can cause negative differences; keep raw values
# in the summary and use clipped values only for the daily chart.
daily = data.diff().fillna(data.iloc[0])
daily.columns = ["Daily Confirmed", "Daily Deaths", "Daily Recovered"]

print("Latest available date:", data.index.max().date())
print("\\nLatest cumulative global totals:")
print(data.iloc[-1].round(0).astype("int64"))

print("\\nFirst five rows of cumulative data:")
print(data.head())

# Save processed data
data.to_csv("covid19_global_cumulative.csv", index_label="Date")
daily.to_csv("covid19_global_daily_changes.csv", index_label="Date")

# Chart 1: cumulative confirmed cases, deaths, recoveries
```

```
plt.figure(figsize=(11, 6))
plt.plot(data.index, data["Confirmed"], label="Confirmed")
plt.plot(data.index, data["Deaths"], label="Deaths")
plt.plot(data.index, data["Recovered"], label="Recovered")
plt.title("Global COVID-19 Cumulative Totals")
plt.xlabel("Date")
plt.ylabel("Reported people")
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig("covid19_cumulative_trends.png", dpi=150)
plt.show()

# Chart 2: daily increases (clipped at zero for easier visualization)
daily_for_chart = daily.clip(lower=0)
plt.figure(figsize=(11, 6))
plt.plot(daily_for_chart.index, daily_for_chart["Daily Confirmed"],
         label="Daily confirmed")
plt.plot(daily_for_chart.index, daily_for_chart["Daily Deaths"],
         label="Daily deaths")
plt.plot(daily_for_chart.index, daily_for_chart["Daily Recovered"],
         label="Daily recovered")
plt.title("Global COVID-19 Daily Reported Changes")
plt.xlabel("Date")
plt.ylabel("Daily reported change")
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig("covid19_daily_changes.png", dpi=150)
plt.show()

print("\\nSaved files:")
print("- covid19_global_cumulative.csv")
print("- covid19_global_daily_changes.csv")
print("- covid19_cumulative_trends.png")
print("- covid19_daily_changes.png")
```

5. Code Explanation

Load data: pandas reads the public CSV files containing cumulative confirmed cases, deaths, and recoveries by location.

Aggregate globally: The code sums the date columns across all listed locations to create worldwide daily time series.

Calculate daily changes: diff() subtracts the previous cumulative value from the current value. Historical data revisions may produce negative differences.

Create charts: Matplotlib plots cumulative totals and daily reported changes so trends can be compared visually.

Export results: to_csv() saves processed tables so they can be opened later in Excel or other analysis tools.

6. Sample Output

The exact totals depend on the source data available when the code is run. Example of the output format:

```
Latest available date: YYYY-MM-DD

Latest cumulative global totals:
Confirmed    <latest reported total>
Deaths       <latest reported total>
Recovered    <latest reported total>

First five rows of cumulative data:
Date        Confirmed    Deaths    Recovered
...         ...          ...       ...

Saved files:
- covid19_global_cumulative.csv
- covid19_global_daily_changes.csv
- covid19_cumulative_trends.png
- covid19_daily_changes.png
```

These are placeholders showing the output structure, not live totals. Run the script to obtain current values from the selected historical dataset.

7. Files Generated

• covid19_global_cumulative.csv — cumulative worldwide totals by date.

• covid19_global_daily_changes.csv — daily changes calculated from cumulative totals.

• covid19_cumulative_trends.png — chart of cumulative confirmed cases, deaths, and recoveries.

• covid19_daily_changes.png — chart of daily reported changes.

8. Project Report

Title: COVID-19 Data Analysis Using Python

Introduction: COVID-19 data analysis helps summarize how reported cases, deaths, and recoveries changed over time. This project uses Python to process public historical data and display global trends.

Methodology: Three CSV time-series files are loaded. Values are aggregated across locations, combined into one table, and used to calculate daily changes. Line charts are generated, and cleaned tables and charts are saved as files.

Results: The program prints the latest date and totals available in the source files and creates charts showing cumulative totals and daily reported changes. Results depend on the dataset snapshot and its reporting revisions.

Conclusion: This project demonstrates data loading, cleaning, aggregation, time-series calculations, visualization, and export using Python. Reported historical data can be revised and should be interpreted with care; charts show reported figures rather than all infections.

9. Troubleshooting

• Connection error: check internet access and try again later.

• A source CSV is unavailable: the upstream repository may have changed or stopped updating a file. Use an available archived dataset and update the URL.

• ModuleNotFoundError: install the missing package with pip.

• Charts do not appear: look for the PNG files saved in the working directory.